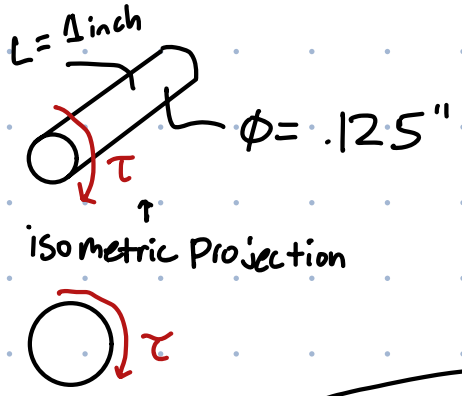
A8: Gear Box

FBD



isometric projection

Knowns

dia = .125 in

L = 1 in

Unknowns

Sheer Stress

or

Torque

$$\tau = \frac{T \cdot c}{J} \qquad J = \frac{\pi D^4}{32} \qquad c = \frac{d}{2}$$

$$\tau = \frac{(32)(T)(c)}{(\pi)(D^4)}$$

$$\frac{T_{in}}{T_{out}} = \frac{G_{out}}{G_{in}}$$

$$T_{in} = \frac{(G_{out})(T_{out})}{(G_{in})}$$

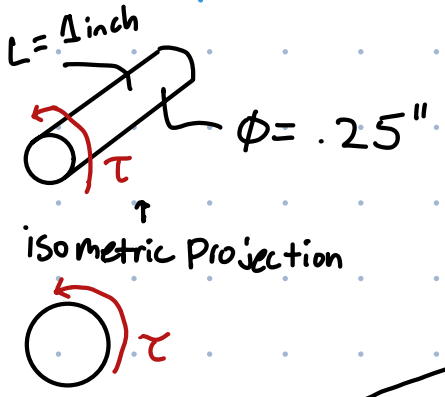
$$\tau = \frac{(32)(G_{out})(T_{out})(c)}{(G_{in})(\pi)(D^4)} \rightarrow \frac{(32)(9)(1.25)(.0625)}{(24)(\pi)(.125)^4} = 1222.309 \text{ psi}$$

$$\frac{(in \cdot lb)(in)}{in^4} \rightarrow \frac{lb}{in^2} = psi$$

Yes, the input shaft is at a sufficient thickness based upon sheer stress.

Out Put Shaft

FBD



Knowns:

dia = .25 in

L = 1 in

T_{out} = 1.25 in-lbf

Unknowns:

Sheer Stress

or

Torque

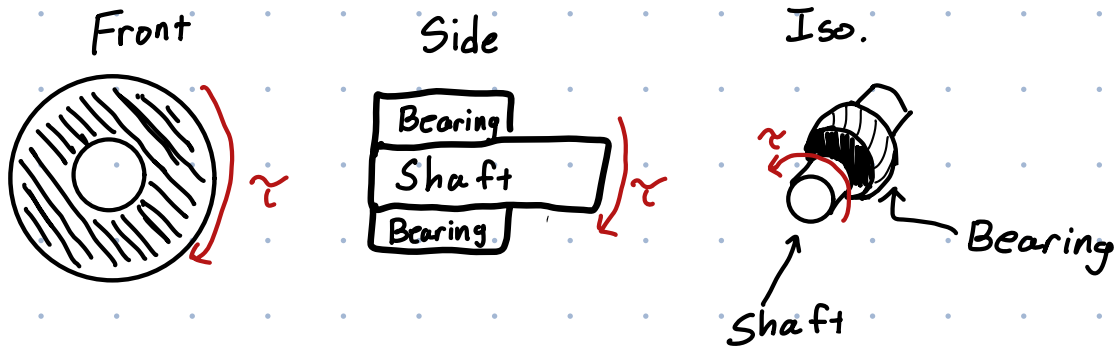
$$\tau = \frac{T \cdot c}{J} \quad J = \frac{\pi D^4}{32} \quad c = \frac{d}{2}$$

$$\tau = \frac{(32)(T)(c)}{(\pi)(D^4)} \longrightarrow \frac{(32)(1.25)(.125)}{(\pi)(.25)^4} = 407.437 \text{ Psi}$$

$$\frac{(\text{in-lb})(\text{in})}{\text{in}^4} = \frac{\text{lb}}{\text{in}^2} = \text{Psi}$$

Yes, the output Shaft is at a sufficient thickness based upon Sheer Stress.

Jornel Bearing



Known:

$RPM_{out} = 1100$

$Torque_{out} = 1.25$ in-lbf

dia = .125 in

$G_r = \frac{24}{9}$ $\quad \alpha = 20°$

Unknowns:

RPM_{in} $\quad$ Area

$Velocity_{in}$, $Velocity_{out}$

$Force_{radial}$ $\quad$ $Force_{tangential}$

$Torque_{in}$

$$RPM_o = \frac{RPM_{in}}{G_r} = (RPM_o)(G_r) = RPM_{in} \qquad V = RPM \cdot \pi \cdot d$$

$$\tau_{in} = \frac{G_{out}}{G_{in}}(\tau_{out}) \qquad F_r = F_t \cdot \tan(\alpha), \quad F_t = \frac{2\tau}{d}$$

in-lb

$$\tau_{in} = \frac{9}{24}(1.25) = .469 \text{ in-lb} \qquad F_{to} = \frac{2(1.25)}{.125} = 20 \text{ lb}$$

in

$$F_{ro} = 20 \cdot \tan(20) = 7.279 \text{ lb}$$

$$F_{ti} = \frac{2(.469)}{.25} = 3.752 \text{ lb}, \quad F_{ri} = 3.752 \cdot \tan(20) = 1.366 \text{ lb}$$

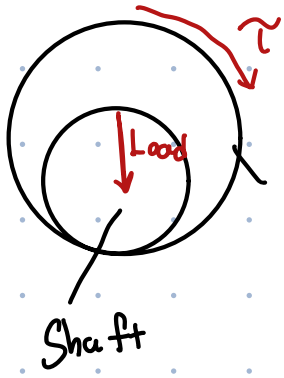
$$V_{in} = \left(1100 \frac{\text{Rev}}{\text{min}}\right) \cdot \left(\frac{24}{9}\right) \cdot (\pi) \cdot (.125 \text{ in}) = 1151.917 \text{ in/min} \text{ or } 95.993 \text{ ft/min}$$

$$V_{out} = (1100) \cdot (\pi) \cdot (.25) = 863.938 \frac{\text{in}}{\text{min}} \text{ or } 71.995 \text{ ft/min}$$

$P_{out} = \frac{7.279}{(.25)(.25)} = 116.464 \text{ psi} \rightarrow PV = 116.469 \cdot 71.995 = 8384.826$

$P_{in} = \frac{1.366}{(.25)(.125)} = 43.712 \text{ psi} \rightarrow PV = 43.712 \cdot 95.993 = 4196.046$

Material: PTFE / RTFE



Bearing
(not to scale)

Knowns

$S_y = 2000$ psi
$P_{in} = 43.712$ psi
$P_{out} = 116.464$ psi
$d_{in} = .125$
$d_{out} = .2$
$F_{ri} = 1.366$ lb
$F_{ro} = 7.279$ lb
$L_B = .25$ in

Unknowns

T_i
T_o

$S_y > \frac{F}{A}$

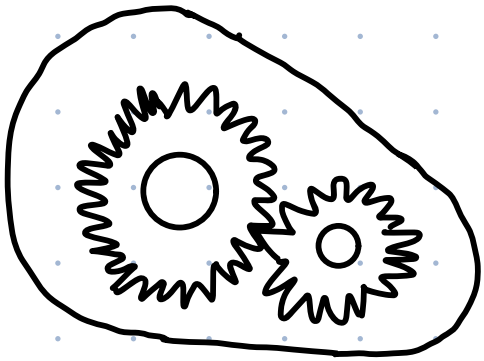
$T > \frac{F}{2\pi rL}$

$T_i = \frac{1.366}{2(2000)(1.25)}$ $\quad T_o = \frac{7.279}{2(2000)(.25)}$

$\Rightarrow 1.366 \times 10^{-3}$ in $\quad \Rightarrow 7.279 \times 10^{-3}$ in

$T_i > 1.366 \times 10^{-3}$ in & $T_o > 7.279 \times 10^{-3}$ in

Front



Isometric

